

Test of Mathematics for University Admission
Mock Set F Paper 1
 Applications of Mathematical Knowledge

Time allowed: **75 minutes**
 Questions: 20 multiple choice
 Calculators: not permitted
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Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Every question uses AS content only. The difficulty lies in how that content is used: parameters that change the case, conditions that quietly remove a solution, and answers that come from structure rather than from brute force.

[illegible]

Question 1

Given that $a = \log_2 3$ and $b = \log_3 5$, express $\log_{15} 20$ in terms of a and b .

- A $\frac{2+ab}{a+b}$
- B $\frac{2+ab}{a(1+b)}$
- C $\frac{2+b}{a+ab}$
- D $\frac{1+2ab}{a(1+b)}$
- E $\frac{2a+b}{a+b}$
- F $\frac{2+a}{1+ab}$

Question 2

Which of the following single transformations map the curve $y = 4^x$ onto the curve $y = 2^{2x+3}$?

- I A translation by $\frac{3}{2}$ in the negative x -direction.
 - II A stretch parallel to the y -axis with scale factor 8.
 - III A stretch parallel to the x -axis with scale factor $\frac{1}{2}$.
 - IV A translation by 3 in the positive y -direction.
- A I only
 - B II only
 - C I and II only
 - D I and III only
 - E II and IV only
 - F I, II and III only

Question 3

The geometric series $3 + 6 + 12 + 24 + \dots$ is summed term by term. What is the smallest number of terms for which the sum exceeds 1 000 000?

- A 17
- B 18
- C 19
- D 20
- E 21
- F 22

Question 4

Find the coefficient of x^3 in the expansion of $(1 + x + x^2)^5$.

- A 10
- B 15
- C 20
- D 25
- E 30
- F 35

Question 5

Let $r(x)$ be the remainder when x^{100} is divided by $x^2 - 3x + 2$. Find $r(0)$.

- A $2^{100} - 2$
- B 1
- C $2^{100} - 1$
- D $2 - 2^{100}$
- E 0
- F 2^{99}

Question 6

For how many real values of k does the equation $x^3 - 3x^2 = k$ have exactly two distinct real solutions?

- A 0
- B 1
- C 2
- D 3
- E 4
- F infinitely many

Question 7

Find the total measure, in degrees, of the set of x in $0^\circ \leq x \leq 360^\circ$ for which

$$2 \cos^2 x > 3 \sin x.$$

- A 60°
- B 120°
- C 180°
- D 240°
- E 300°
- F 330°

Question 8

The equation $x^2 + (k - 3)x + k = 0$ has real roots α and β with $\alpha^2 + \beta^2 = 7$. Find all possible values of k .

- A $k = 4 - \sqrt{14}$ only
- B $k = 4 + \sqrt{14}$ only
- C $k = 4 \pm \sqrt{14}$
- D $k = 1$ or $k = 9$
- E there are no such k
- F $k = 4 - \sqrt{7}$ only

Question 9

Find the complete set of real values of x for which

$$\log_{1/2}(x^2 - 3x) > -2.$$

- A $-1 < x < 4$
- B $x < -1$ or $x > 4$
- C $-1 < x < 0$ or $3 < x < 4$
- D $0 < x < 3$
- E $x < 0$ or $x > 3$
- F $3 < x < 4$

Question 10

How many solutions does the equation $\tan 3x = \tan x$ have in the interval $0^\circ \leq x \leq 360^\circ$?

- A 1
- B 3
- C 4
- D 5
- E 7
- F 9

Question 11

A circle lies in the first quadrant, touches both coordinate axes, and also touches the line $3x + 4y = 12$. Find the sum of all possible radii.

- A 1
- B 2
- C 3
- D 5
- E 6
- F 7

Question 12

In triangle ABC , $AB = 5$ and $AC = 7$, and the median from A to the midpoint M of BC has length 4. Find BC .

- A $\sqrt{21}$
- B $2\sqrt{21}$
- C $2\sqrt{10}$
- D 6
- E $\sqrt{74}$
- F 8

Question 13

How many real values of k are there for which the region enclosed by the curve $y = x^2$ and the line $y = kx$ has area 36?

- A none
- B exactly one
- C infinitely many
- D exactly two
- E exactly three
- F exactly four

Question 14

The curve $y = x^3 + px^2 + qx$ has a stationary point at $(1, -4)$. Find the y -coordinate of its other stationary point.

- A -4
- B $\frac{44}{27}$
- C 13
- D $\frac{49}{3}$
- E $\frac{392}{27}$
- F $-\frac{392}{27}$

Question 15

Find the area of the region of points (x, y) satisfying all three of

$$y \geq x^2, \quad y \leq 2x + 3, \quad x \geq 0.$$

- A $\frac{5}{3}$
- B $\frac{16}{3}$
- C 7
- D 8
- E 9
- F $\frac{32}{3}$

Question 16

The curves $y = x^2$ and $y = -(x - 3)^2$ have two common tangent lines. Find the point where these two lines meet.

- A $(\frac{3}{2}, 0)$
- B $(3, 9)$
- C $(0, -9)$
- D $(\frac{3}{2}, -\frac{9}{2})$
- E $(0, 0)$
- F $(3, 0)$

Question 17

A cylinder is inscribed in a sphere of radius 3. Find the greatest possible volume of the cylinder.

- A $12\sqrt{3}\pi$
- B 36π
- C 24π
- D $8\sqrt{3}\pi$
- E 16π
- F $18\sqrt{2}\pi$

Question 18

Find the sum of all the integers from 1 to 500 inclusive that are divisible by 3 or by 7, but not by both.

- A 5796
- B 36 015
- C 41 583
- D 47 883
- E 53 679
- F 59 475

Question 19

Find the minimum value of

$$\frac{x^2 + 3}{\sqrt{x^2 + 2}}$$

over all real x .

- A $\frac{3}{2}$
- B $\sqrt{3}$
- C 2
- D $\frac{3\sqrt{2}}{2}$
- E $\frac{5}{2}$
- F $2\sqrt{2}$

Question 20

For how many integers c does the equation

$$x^3 - 6x^2 + 9x + c = 0$$

have three distinct positive real roots?

- A 1
- B 2
- C 3
- D 4
- E 5
- F 6

END OF PAPER